



Republic of Rwanda
Ministry of Education



HYDROGRAPHIC SURVEYING

LSVHS501

Conduct Hydrographic Surveying

Competence

RQF Level: 5

Learning Hours



Credits: 7

Sector: Construction and Building Services

Trade: Land Surveying

Module Type: Specific

Curriculum: CBSLSV5002-TVET Certificate V in Land Surveying

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Purpose statement	This module describes the skills, knowledge and attitude required to conduct hydrographic surveying. It enables learners to plan hydrographic surveying activities, to collect hydrographic surveying data and to produce nautical and bathymetric charts.					
Learning assumed to be in place	Apply Remote Sensing					
Delivery modality	Training delivery		100%	Assessment		Total 100%
	Theoretical content		30%	Formative assessment	30%	50%
	Practical work:		70%		70%	
	• Group project and presentation	20%				
	• Individual project /Work	50%				
			Summative Assessment		50%	

Elements of Competency and Performance Criteria

Elements of competency	Performance criteria
1. Plan hydrographic surveying activities	1.1. Site Reconnaissance is properly conducted according to the environmental conditions and specific procedures
	1.2. Working methodologies are properly identification according to the type of project
	1.3. Resources are properly allocated according to the project durationn and Project activities
2. Collect hydrographic surveying Data	2.1. Hydrographic data are properly identified according to their types and purpose
	2.2. Control points are accurately established according to the site conditions.
	2.3. Hydrographic data collection methods are correctly applied according to their data use
3. Produce hydrographic charts	3.1. Hydrographic charts elements are well identified according to project specifications
	3.2. Hydrographic charts are correctly generated based on their use
	3.3. Hydrographic charts are properly published according to their applications.

Intended Knowledge, Skills and Attitude

Knowledge	Skills	Attitude
✓ Describing work planning procedures ✓ Describing hydrographic Surveying principles ✓ Describe hydrographic surveying procedures ✓ Identifying hydrographic surveying tools, equipment and materials ✓ Identifying nautical and bathymetry charts elements ✓ Identifying geodetic controls networks ✓ Describing cartographic principles ✓ Identifying hydrographic data analysis methods ✓ Describe safety measures principles in hydrographic surveying profession ✓ Describing principles of ethics and regulatory compliance	✓ Designing work plan ✓ Operating hydrographic surveying instruments/equipment ✓ Performing accurate data collection ✓ Applying data management skills ✓ Applying Hydrographic data analysis techniques ✓ Applying surveying mathematics ✓ Using Surveying software ✓ Creating nautical and bathymetric charts ✓ Applying publishing skills ✓ Applying safety precaution ✓ Applying communication skills ✓ Applying Surveying mathematics ✓ Operating hydrographic surveying equipment ✓ Applying reporting skills ✓ Applying managerial skills ✓ Applying communication skills ✓ Applying ICT skills	✓ Embracing continuous learning and improvement ✓ Showing adaptability to changing environments ✓ Being decision maker ✓ Being attentive ✓ Being creative and innovative ✓ Having problem solving attitude ✓ Having teamwork spirit ✓ Having time management attitude ✓ Being Flexible ✓ Being Patient ✓ Being responsible ✓ Being persistent ✓ Being honest ✓ Being Self-motivated ✓ Being research oriented ✓ Being attentive ✓ Being Detail-oriented ✓ Having Critical Thinking attitude ✓ Being prompt ✓ Being confident

Course content

Learning outcomes	At the end of the module the learner will be able to: 1. Plan hydrographic surveying activities 2. Collect hydrographic surveying data 3. Produce hydrographic charts
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Learning outcome 1: Plan hydrographic surveying activities	Learning hours: 15 hours
Indicative content	
• Conducting hydrographic surveying field reconnaissance ✓ Preparing field reconnaissance ✓ Identifying the scope of work  Specifications of hydrographic surveying  Hydrographic survey project boundaries  Hydrographic survey project outline ✓ Conducting site visit • Identifying hydrographic surveying methods ✓ Control surveying methods ✓ Topographic survey at coast ✓ Bathymetric survey methods • Allocating hydrographic surveying resources ✓ Identifying hydrographic surveying activities ✓ Selecting tools and Equipment ✓ Selecting human resources ✓ Estimating time required	
Resources required for the indicative content	
Equipment	GNSS Receivers, Echo Sounder dual channel, Automatic tide gauge, Side Scan sonar, Boat, Laptop, White/Black Board
Materials	Markers, dusters, chalks

Tools	Velocity Meters, Bottom samplers, Internet, Thermometer
Facilitation techniques	Group discussion, Brainstorming, Presentations, Pictures/Drawings, scenario, individual activity - Trainer shows pictures and displays video of some water bodies and surveying activities, and in manageable groups, he/she asks the learners to discuss on location insights, and scope of the work. - Trainer provides a scenario of collecting site information and in manageable groups, students discuss and present the report of their findings. Trainer asks the students in small groups to discuss hydrographic surveying methods. - Trainer asks learners to allocate resources based on the findings from the scenario.
Formative assessment methods	Written assessment, Oral presentation, Performance assessment

Learning outcome 2: Collect hydrographic surveying data	Learning hours: 30 hours
Indicative content	
Identifying hydrographic surveying data ✓ Ashore surveying data ✓ Bathymetric surveying data ✓ Nautical surveying data Establishing Control Points ✓ Identifying Control points standards ✓ Establishing Horizontal Control ✓ Establishing vertical control Applying hydrographic surveying data collection methods ✓ Staff method ✓ Rope method ✓ Satellite positioning method ✓ Echo Sounding ✓ Side scanning ✓ Airborne (LiDAR) method	

Resources required for the indicative content

Equipment	Vessel, Multibeam and scanning Sonars, Autonomous Underwater Vehicles (AUVs), Cameras, Inertial Navigation Systems, Sound Velocity Sensors, GPS/GNSS, Laser scanner, Workshop resources, Projector, Lap top
Materials	White/Black Board, Markers, chalks
Tools	3D Mechanical Scanning Software, Video aid, Internet
Facilitation techniques	Scenarios, trainer guided, video show • Trainer displays hydrographic survey data through video show, and asks learners to observe and identify different hydrographic data. • Trainer provides a scenario about hydrographic surveying and guides the learners to establish the control points. • Trainer shows step by step how to collect hydrographic data using actual equipment and tools. The trainer uses video demonstration to provide detailed instructions and guidance.
Formative assessment methods	• Written assessment, • Oral presentation • Performance assessment

Learning outcome 3: Produce hydrographic charts

Learning hours: 25 hours

Indicative content

- **Identifying elements of hydrographic charts**
 - ✓ Elements of bathymetric chart
 - ✓ Elements of nautical chart
- **Generating hydrographic charts**
 - ✓ Download and upload data
 - ✓ Process the data
 - ✓ Analyze and interpret data
 -  Generation of Digital Elevation Models
 -  Generation of depth contours
 - ✓ Produce nautical chart
 - ✓ Produce bathymetric chart

- **Publishing hydrographic charts**

- ✓ Customizing the chart layout
- ✓ Exporting hydrographic charts
- ✓ Plotting hydrographic charts

Resources required for the indicative content

Equipment	Computer, Workstation, Projector
Materials	Reference Charts, Satellite imagery, Aerial photography, Hydrographic survey data, Basemaps.
Tools	Flip chart, Flip chart stand, drafting tools, GIS Software, ENVI software, ERDAS Imagine software Internet, Marker
Facilitation techniques	Group discussion, Presentations, Mapping, individual activity • Trainer displays a video or drawings on hydrographic charts. In manageable groups, Trainer asks learners to identify types and elements of hydrographic charts. • Trainer provides layout and creates a scenario of hydrographic chart and asks learners to produce and publish hydrographic charts.
Formative assessment methods	• Written assessment, • Oral presentation • Performance assessment • Product assessment

Integrated situation

Residents of Ngoma and Bugesera Districts are facing the problem of exchanging crops crossing Mugesera Lake. The Ministry of Agriculture (MINAGRI) has initiated a plan to improve marine transport means through the lake to facilitate exchange of crops and interlinkage of the neighboring districts. In collaboration with the two Districts, the above institution would like to undertake the by-water ways project within Mugesera lake.

In this framework, MINAGRI is looking for a competent company specialized in hydrographic surveying works to produce a hydrographic chart of Mugesera Lake and ABC Engineering Ltd is hired for the task.

The required chart should indicate the following features:

- ✓ The boundary of Mugesera Lake
- ✓ Topographic features along the coast
- ✓ Hydrographic chart elements

As a hydrographic surveyor of ABC Engineering Ltd, you are requested to perform this task within eight (8) working hours. All needed related data will be collected from Ministry of Agriculture (MINAGRI) while related equipment should be provided by the contractor.

Resources

Tools	Handheld, Velocity Meters, Bottom samplers, Internet, Thermometer
Equipment	Vessel, Multibeam and scanning Sonars, Autonomous Underwater Vehicles (AUVs), Cameras, Inertial Navigation Systems, Sound Velocity Sensors, GPS/GNSS, Laser scanner Computer, Workstation
Materials/ Consumables	Reference Charts, Satellite imagery, Aerial photography, Hydrographic survey data, Base maps.

Assessable outcomes	Assessment criteria (Based on performance criteria)	Indicator	Observation		Marks allocation
			Yes	No	
Learning outcome 1: Plan hydrographic surveying activities	Site Reconnaissance is well conducted	Field reconnaissance is prepared			2
		Scope of the work is identified			4
		Site visit is done			3
	Working methodology is properly identified	Control points are identified			3
		Ashore topographic surveying method is identified			3
		Nautical surveying methods are selected			2
	Resources are well allocated	Hydrographic surveying activities are identified			3

		Tools are selected			2
		Equipment is selected			3
Learning outcome 2: Collect hydrographic surveying Data	Data collection methods are properly selected	Ashore topographic data are identified			5
		Nautical surveying data are identified			5
	Control points are accurately established	Control points standards are identified			5
		Control points are established			5
	Hydrographic surveying data are correctly captured	Staff method is applied			5
		Rope method is applied			5
		Satellite positioning method is applied			5
Learning outcome 3: Produce hydrographic chart	Nautical chart elements are well identified	Elements of nautical chart are identified			8
	Nautical chart is correctly created	Hydrographic surveying models are generated (Contours, DEM)			8
		Nautical chart is produced			6
	Nautical chart is properly published	Nautical chart layout is customized			8
		Nautical chart is exported			5
		Nautical chart is plotted			5
Total marks		100			
Percentage Weightage		100%			
Minimum Passing line % (Aggregate): 70%					

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